

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 30 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 30 trials
Total training paths	15 · 454 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports/Configuration10

Sensor selection

Hands	right, left
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	32 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Sensor grid (✓ enabled, ✗ disabled, — not present)

Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✗	✗	✓	✓
left index	✗	✗	✓	✓
left middle	✗	✗	✓	✓
left ring	✗	✗	✓	✓
left pinky	✗	✗	✓	✓
left wrist	✗	—	—	—
right palm	✓	—	—	—
right thumb	✗	✗	✓	✓
right index	✗	✗	✓	✓
right middle	✗	✗	✓	✓
right ring	✗	✗	✓	✓
right pinky	✗	✗	✓	✓
right wrist	✗	—	—	—

Enabled: 22 · Disabled: 22 (channels per enabled IMU depend on modality flags above).

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	robust
Feature mode	stats+fft → feature dim 1728
Test size · seed · CV folds	0.1 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=2040 · test=46

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

CV = 5-fold cross-validation. Test = hold-out accuracy. M-* = macro-averaged, W-* = support-weighted. Prec / Rec / F1 on hold-out test.

Algorithm	CV acc	CV std	CV F1	CV F1 σ	Test	M-Prec	M-Rec	M-F1	W-Pre c	W-Rec	W-F1
Random Forest	1.0000	0.0000	1.0000	0.0000	0.7391	0.7516	0.7381	0.7405	0.7517	0.7391	0.7409

Per-class metrics (hold-out test)

=== Random Forest ===

	precision	recall	f1-score	support
Double_Closed_Fist	0.75	0.75	0.75	8
Double_Okay	0.55	0.75	0.63	8
Double_Open_Palm	0.50	0.43	0.46	7
Double_Phone	0.71	0.62	0.67	8
Double_Pistol	1.00	0.88	0.93	8
Double_Spiderman	1.00	1.00	1.00	7
accuracy			0.74	46
macro avg	0.75	0.74	0.74	46
weighted avg	0.75	0.74	0.74	46

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.1667

Per-class external accuracy

Class	Random Forest	Support
Double_Closed_Fist	0.000	5
Double_Okay	0.000	5
Double_Open_Palm	0.000	5
Double_Phone	0.000	5
Double_Pistol	0.000	5
Double_Spiderman	1.000	5